

Analyses Randomization

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Load libraries

If the libraries are not installed yet, you need to install them using, for example, the command: `install.packages("ggplot2")`.

```
library(ggplot2)
library(ggExtra)
library(ggpubr)
library(gridExtra)
```

Load Data

Load csv file with quantitative feature values for sign sequences per object.

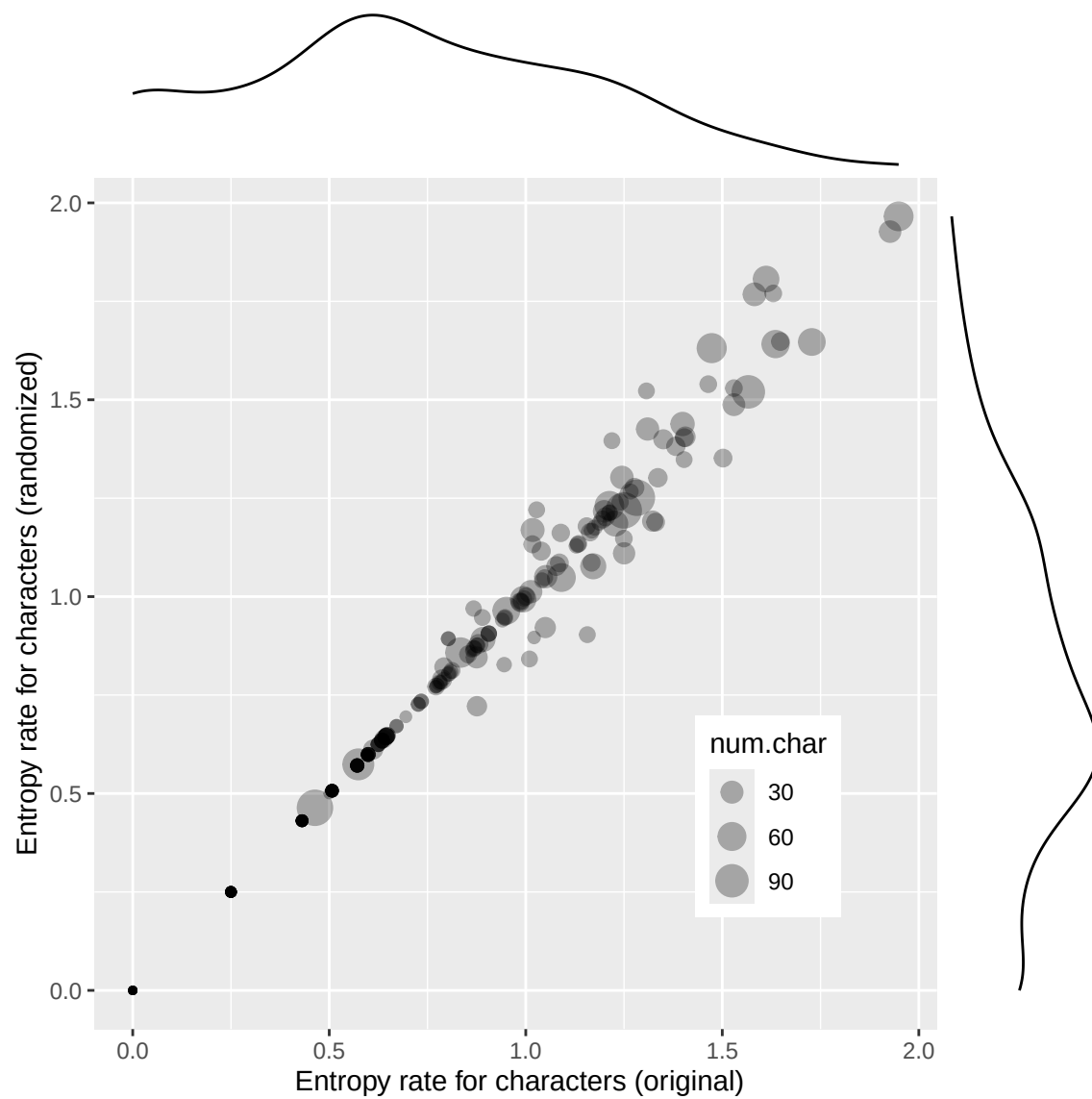
```
# load signbase estimations for strings on objects
sign.data <- read.csv("~/Github/UpperPalCombinatorics/results/signBase_features.csv")
# select Aurignacian objects
sign.data <- sign.data[sign.data$archaeological_unit == "Aurignacien", ]
```

Plot

Entropy rate vs. Unigram entropy

Entropy rate vs. unigram entropy for the original versus randomized sequences.

```
hrate.random.plot <- ggplot(sign.data, aes(x = hrate.chars, y = hrate.chars.random,
                                             size = num.char)) +
  geom_point(alpha = 0.3) +
  theme(legend.position = c(.8, .25)) +
  labs(x = "Entropy rate for characters (original)", y = "Entropy rate for characters (randomized)")
hrate.random.plot <- ggMarginal(hrate.random.plot,
                                type = "density")
hrate.random.plot
```



Save complete figure to file

```
ggsave("~/Github/UpperPalCombinatorics/figures/combinedPlot_signBase_random.pdf",
        hr.rate.random.plot, dpi = 300, width = 6, height = 6, device = cairo_pdf)
```

Correlation

```
cor(sign.data$hr.rate.chars, sign.data$hr.rate.chars.random, method = "pearson")
```

```
## [1] 0.9937936
```